Kaitaro Hori

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EDUCATION

University of California, Davis Graduation: June 2026
Bachelor of Science, ***Aerospace Science and Engineering*** GPA: 3.61/4.0
Bachelor of Science, ***Mechanical Engineering***

- Dean's Honors List, Fall 2022
- Relevant Coursework: Fluid Mechanics, Aircraft Propulsion, Applied Aircraft Aerodynamics, Thermodynamics, Heat Transfer, Mechanics of Materials, Stability and Control of Aerospace Vehicles, Automatic Control of Engineering Systems, Orbital Mechanics, Rocket Propulsion, Space Vehicle Design, Compressible Aerodynamics

SKILLS

Computer: Microsoft Word, PowerPoint, LaTeX, Google Docs, Excel
Technical Skills: Proficient: MATLAB, Simulink, Intermediate: Python, XFoil, C#, Unity, SOLIDWORKS, OnShape, Fusion, FreeFlyer, NASTRAN/PATRAN, OpenVSP, Ansys Fluent, Milling, Lathing, CNC Milling
Languages: Japanese (Fluent), English (Fluent)

RELEVANT EXPERIENCE

Team Member - ***UC Davis Aggie Propulsion Rocketry Lab*** September 2023 - September 2024

- Co-founded a new department focused on control systems for a liquid rocket system.
- Conducted research on system dynamics and dynamic control for a landing system.

RESEARCH EXPERIENCE AND PROJECTS

Research Assistant - ***Human/Robotics/Vehicle Integration & Performance Laboratory*** Feb 2026 - Present

- Contributing to the Mission Operations and Observation Nexus (MOON) initiative under Prof. Robinson by developing ground-station capabilities, including QFH antenna systems and autonomous satellite tracking, reception, and image-processing workflows for LEO and GEO satellites.
- Supported the design of a university mission-operations facility for real-time satellite data reception, decoding, and analysis in support of education and mission-simulation operations.

Research Intern - ***NeuralX*** July 2025 - September 2025

- Designed Smoothed Particle Hydrodynamics (SPH) simulations to model fluid and water-like behavior
- Optimized compute shader performance in Unity and integrated numerical methods. Explored advanced simulation techniques and documented findings to support ongoing R&D in physics-based modeling.

Research Assistant - ***UC Davis Computational Flow Physics and Aeroacoustics Lab*** December 2024 - April 2025

- Collaborated in a team of two to conduct a computational research project in aeroacoustics using NASA's ANNOP2.
- Worked as a team to solve and comprehend the complex nature of the software on Microsoft VS in Fortran.

Payload Lead - ***Spacecraft Inspection Mission Redesign Project*****,** ***Senior Capstone Class*** January 2026 - March 2026

- Worked as a system/payload engineer to redesign the NASA CPOD CubeSat proximity-operations mission into an autonomous ISS inspection concept focused on close-range, non-docking operations.
- Developed a feasible inspection-oriented mission architecture by evaluating LiDAR integration, sensor tradeoffs, and onboard processing needs to improve RPO, inspection capability, and mission safety.

Communication Lead - ***Spacecraft Inspection Mission Design Project, Senior Capstone Class*** April 2026 -Present

- Responsible for preliminary communications architecture, antenna and link-budget sizing, and downlink/TT&C trade studies in coordination with power and payload teams for a heliocentric-orbit satellite inspection mission.
- Led Communications, C&DH, and Ground Control System design by developing data budgets, mission-mode downlink timelines, contact-window analyses, onboard data-handling assumptions, and ground-station architecture.

OTHER WORK EXPERIENCE

Tutor - ***UC Davis Academic Assistance and Tutoring Centers*** April 2025 - Present

- Provide academic assistance to undergraduate students in Electromagnetism, Calculus, Dynamics, and Linear Algebra through both drop-in and individual tutoring sessions.

Student Assistant - ***UC Davis International Center*** June 2024 - Present

- Assisted exchange students with their English as needed from a bilingual perspective as a teaching assistant.
- Utilized intercultural communication skills and served as a role model for international community members.
- Worked as a course interpreter for Japanese trainees from the Japanese Agricultural Training Program in partnership with the Japanese Agricultural Exchange Council.